

FGE-ECO-REGISTRY-v2.0

Doc ID: FGE-ECO-REGISTRY-v2.0 **Layer:** L2 (Engine) — sits above Character Suite, feeds the Foundry
Status: DRAFT — pending lock under LAW-NORM-002 **Supersedes:** FGE ECOSYSTEM PRESET
REGISTRY v1.0 (deprecated — flat preset list, no schema) **Precedence:** Subordinate to Living Canon
Foundation (FGE-FOUND-LIVING-001) and the Invariant Kernel. **The kernel always wins.**

0. Why v1.0 Is Deprecated

v1.0 modeled ecosystems as *bundles*: each preset hard-coded architecture + myth density + wealth + character mods into one opaque prose blob. This fails on three axes that matter to a render pipeline:

1. **Not composable.** You cannot blend “Dragon Dynasty” with “Ultra Modern,” diff two ecosystems, or interpolate between them. A bundle is a dead end.
2. **Not deterministic.** Prose attributes (“subtle divine proportions”) cannot be validated, hashed, or version-pinned. They leak into prompts as improvisation — a direct violation of *derive, never improvise*.
3. **No precedence.** v1.0 silently lets an ecosystem bolt scale accents or “enhanced genetics” onto a character. That collides with the five locked kernel fields with no rule to arbitrate. It fights LAW-NORM-002/003 by omission.

v2.0 replaces the bundle with a **coordinate**: an ecosystem is a named binding of independent axis values plus a *constraint envelope*, never a payload.

1. Core Principle

The ecosystem proposes. The canon disposes.

An ecosystem may set the world (backdrop, tech, mood) and may *request* character modifications — but only inside a character’s Open Evolution Space. It can never touch the Invariant Kernel. This is enforced at the schema level (Section 4) and gated before render (Section 6), not left to operator discipline.

2. Axis Decomposition

Every ecosystem is expressed as values on independent axes. Axes are orthogonal: changing one never implies changing another.

Axis	Type	Range / Domain	Notes
<code>setting_id</code>	enum (FK)	Setting Registry (§3)	The renderable backdrop: architecture, environment, clothing logic, environmental storytelling.
<code>tech_level</code>	enum	<code>ancient</code> , <code>historical</code> , <code>industrial</code> , <code>near_future</code> , <code>post_scarcity</code> , <code>collapsed</code>	Decoupled from setting so “ruined + post-scarcity” (Dust & Gold) is expressible.
<code>myth_density</code>	scalar	0–5	Was a star rating. Now a real knob the Foundry reads.
<code>wealth_level</code>	scalar	0–5	Distribution ceiling.
<code>poverty_contrast</code>	scalar	0–5	Separate scalar. v1.0 conflated wealth and contrast; Dust & Gold needs both maxed independently.
<code>mood_vector</code>	enum[]	controlled vocabulary	e.g. <code>noble</code> , <code>hedonistic</code> , <code>seductive</code> , <code>brutal</code> . Max 3, ranked.
<code>mod_envelope</code>	object	§4	The constraint, not a payload. The dangerous axis, made safe.

Why this is the n+3 move: a flat list of 10 presets is n+1. Grouping them into super-families (v1.0's closing table) is n+2 — taxonomy without mechanism. Decomposing into orthogonal axes with a constraint envelope and a handoff contract is n+3: the 10 presets become 10 *coordinates* in a continuous, composable, validatable space, and new ecosystems become free (any valid axis binding is an ecosystem).

3. Setting Registry (decoupled backdrop layer)

`setting_id` references a separate, render-only table. Settings carry **zero** character logic — purely backdrop. This is what makes blends sane: you blend mood/myth/wealth scalars freely while settings stay discrete and swappable.

setting_id	Backdrop summary
FGE-SET-marble-civic	Marble megacities, golden civic architecture, floating monuments
FGE-SET-ultra-modern	Penthouses, hypercars, yachts, fashion districts
FGE-SET-ruined-luxury	Luxury compounds amid ruined civilization, desert highways
FGE-SET-imperial-solar	Solar-system civilization, imperial capitals

FGE-SET-ocean-crystal	Ocean megacities, crystal technology
FGE-SET-arena-empire	Arenas, temples, caravans
FGE-SET-sky-citadel	Dragon ports, aerial commerce, sky citadels
FGE-SET-frozen-rune	Frozen megacities, rune technology, cyber longships
FGE-SET-old-money	Old-money estates, cathedral architecture
FGE-SET-stellar-frontier	Frontier worlds, asteroid ranches, trade hubs

4. The `mod_envelope` (constraint, never payload)

This is the field that makes v2.0 safe. An ecosystem cannot *write* a character mod; it can only declare which mods are *permitted to be requested* inside Open Evolution Space, and it explicitly cannot override the kernel.

```
"mod_envelope": {
  "permitted": ["rune_jewelry", "ceremonial_armor", "scale_accents"],
  "kernel_locked": true,
  "forbidden_overrides": [
    "sbid",
    "material_fingerprint",
    "frequency_cluster",
    "core_conflict",
    "signal"
  ]
}
```

- `permitted` — mods the ecosystem may *request*. The character's own Open Evolution Space still decides whether to accept.
- `kernel_locked` — **MUST be `true`**. A registry row with `kernel_locked: false` is invalid and rejected at lock.
- `forbidden_overrides` — the five locked kernel fields, restated at the point of use so the gate (§6) has nothing to infer.

Resolution rule: if a `permitted` mod would alter any kernel field for a specific character (e.g. `"scale_accents"` implies a material change for Raven/Obsidian), the request is dropped for that character and a `GATE` warning is emitted. The ecosystem does not get to win that fight.

5. Handoff Contract Patch — `fge.handoff.v1`

The ecosystem layer sits above character creation, so it **MUST** enter the contract explicitly or it

leaks into prompts non-deterministically. Add this block:

```
"ecosystem": {
  "preset_id": "FGE-ECO-002",
  "axes": {
    "setting_id": "FGE-SET-ultra-modern",
    "tech_level": "near_future",
    "myth_density": 1,
    "wealth_level": 5,
    "poverty_contrast": 0,
    "mood_vector": ["hedonistic", "glamorous", "elite"]
  },
  "mod_envelope": {
    "permitted": [],
    "kernel_locked": true,
    "forbidden_overrides": ["sbid", "material_fingerprint", "frequency_cluster", "core_conflict"]
  },
  "blend": {
    "with": null,
    "weight": 0.0
  }
}
```

Blend semantics (mirrors your DNA/Trend 0.7 default): `blend.with` references a second `preset_id`; `weight ∈ [0,1]` interpolates the *scalar* axes (`myth_density` , `wealth_level` , `poverty_contrast`) and merges `mood_vector` (union, re-ranked, truncated to 3). `setting_id` and `tech_level` **do not** interpolate — they snap to the dominant preset (`weight ≥ 0.5`). Discrete fields never blend; only scalars do.

6. Foundry Gate (pre-render validation)

Before any render call, the Foundry MUST run `ECO-GATE` :

Check	Condition	On fail
ECO-GATE-1	<code>kernel_locked == true</code>	HARD STOP — invalid registry row
ECO-GATE-2	every <code>permitted</code> mod $\notin$ character kernel fields	DROP mod for that character, emit warning
ECO-GATE-3	<code>setting_id</code> resolves in Setting Registry	HARD STOP
ECO-GATE-4	all scalars $\in [0,5]$; <code>mood_vector</code> ≤ 3	HARD STOP
ECO-	if <code>blend.with</code> set, both presets exist and	HARD STOP

GATE-5	weight ∈ [0,1]	
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ECO-GATE runs **before** the existing compile→validate→lock→render→release lifecycle’s render stage. An ecosystem that fails the gate cannot reach Grok.

7. The Ten Presets, Restated as Coordinates

Each former bundle is now a single registry row. (Scalars shown; `mod_envelope` omitted for brevity — all `kernel_locked: true`.)

preset_id	name	setting_id	tech	myth	wealth	poverty	mood (ranked)
FGE-ECO-001	Asgardian Renaissance ⁺	marble-civic	historical	5	5	1	noble, aspirational, legendary
FGE-ECO-002	FGE Ultra Modern	ultra-modern	near_future	1	5	0	hedonistic, glamorous, elite
FGE-ECO-003	Dust & Gold	ruined-luxury	collapsed	2	5	5	scarcity, decadent, brutal
FGE-ECO-004	Imperial Solar Empire	imperial-solar	post_scarcity	3	5	2	order, conquest, prestige
FGE-ECO-005	Atlantean Technocracy ⁺	ocean-crystal	post_scarcity	4	5	1	serene, advanced, elite
FGE-ECO-006	Blood & Bronze	arena-empire	ancient	3	3	3	visceral, heroic, raw
FGE-ECO-007	Dragon Dynasty	sky-citadel	historical	5	4	2	majestic, fierce, regal
FGE-ECO-008	Neo-Valhalla ⁺	frozen-rune	near_future	4	4	2	warrior, augmented, mythic
FGE-ECO-009	Eclipse Aristocracy	old-money	industrial	3	5	1	seductive, mysterious, elite
FGE-	Celestial	stellar-	near_future	2	3	3	free, expansive,

ECO-010	Frontier	frontier					adventurous
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† **IP-PROXIMITY FLAG.** “Asgardian,” “Atlantean,” and “Valhalla” carry the same naming-collision risk that produced the SAG-AFTRA incident. Acceptable as *internal* axis labels. **Before any of these becomes a customer-facing drop name, route through MOD-CITATION-INTEGRITY v0.1.** The v1.0 “Influences” lists (Thor, Dune, Succession, etc.) are fine as internal reference but must never surface in output.

8. Open Questions (for lock review)

1. `mood_vector` **vocabulary** — needs a closed controlled list before lock, or it becomes an improvisation vector.
 2. **Blend depth** — v2.0 allows 2-way blends only. Do you want n-way? (Recommend no until one 2-way blend ships.)
 3. **Per-character ecosystem affinity** — should a character carry an allowed-ecosystem whitelist (Raven reads wrong in Ultra Modern), or is that the operator’s call at compile time? Leaning: soft affinity score, not a hard lock.
 4. **Setting Registry ownership** — does `setting_id` become its own L3 subsystem with its own lifecycle, or stay embedded here? At 10 rows, embed. Past ~30, promote.
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9. Lock Checklist

- `mood_vector` controlled vocabulary defined
- All 10 rows pass ECO-GATE-1 through ECO-GATE-4
- Handoff contract patch merged into `fge.handoff.v1` and version-bumped
- IP-proximity names (†) cleared or quarantined to internal-only
- Semantic identity hash assigned per DECREE-NORM-PAYLOAD-001
- Lock recorded under LAW-NORM-002